

Solution Requirements

Date	5 July 2026
Team ID	
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

Functional Requirements (FR) define specific behaviors, functions, or features of the system (What the system should do).

Non-Functional Requirements (NFR) specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can securely register, log in, log out, and verify their identity using email and password, secured with JWT and bcrypt password hashing.	High
2	Authorization levels	Each borrower can only access their own loans, financial profile, and AI history — enforced via JWT-scoped user identity on every request.	High
3	External interfaces	Integrate Google Gemini AI API for settlement negotiation strategy and letter generation, with a rule-based fallback engine when the API is unavailable.	High
4	Transactions processing	System manages loan data (add/view/delete), financial profile updates, settlement predictions, and AI negotiation requests securely.	High
5	Reporting	Generate a live financial health dashboard, debt timeline projections, settlement recommendations, and AI interaction history for each user.	High
6	Business rules, etc.	Implement financial rules such as DTI ratio calculation, EMI ratio, surplus, financial stress level, and settlement priority logic.	High

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
7	Compliance to laws or regulations	<i>Ensure secure handling of sensitive financial data with hashed passwords, JWT-based access control, and no plaintext credential storage.</i>	High
8	Other	<i>AI negotiation letter generation, structured activity logging, 15-second AI request timeout with fallback, and automatic session expiry handling.</i>	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The dashboard and all API responses should load quickly for the borrower.	Dashboard loads in < 2 seconds; Gemini API calls timeout and fall back after 15 seconds.
2	Scalability	The system should handle growth in users, loans, and AI requests without degradation.	Indexed foreign keys across all 5 tables to keep queries fast as data grows.
3	Security & Data Privacy	Ensure data protection and secure handling of sensitive financial information.	JWT authentication, bcrypt password hashing, no plaintext credentials stored.
4	Reliability & Availability	The system should remain functional even if the external AI service is unavailable.	Rule-based fallback engine automatically activates if Gemini API fails or times out.
5	Usability & Accessibility	Provide an intuitive, responsive experience across devices.	Mobile-responsive UI with hamburger navigation at the 860px breakpoint.
6	Other	Ensure maintainability and ease of deployment.	Modular backend (routers/services), 86 passing backend tests, deployed on Render + Vercel.